

How Robust Are the Legatum Prosperity Index Rankings? A Monte Carlo and Sensitivity Analysis of Weighting, Normalization, and Aggregation

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Abstract

The Legatum Prosperity Index is a composite measure that ranks 167 countries by the equal-weighted average of twelve pillar scores. Here, Monte Carlo uncertainty analysis and variance-based sensitivity analysis were performed on the normalization, weighting, and aggregation used in the final step of the composite calculation. The portion of a weight's influence on the composite that can cause a change in rank was also derived. The published scores of the twelve pillars reproduced each published overall score within an error of 5.0×10^{-11} and each published rank exactly. The twelve pillars all shared one dominant principal component that accounted for 74.5 percent of their total variance. Under three different weighting priors, the country rankings were found to be robust overall, with a median Spearman correlation of at least 0.9899 with the published ranking. However, robustness of ranking did not extend to the middle of the ranking list. Under the uniform prior, the median 90 percent rank interval across all 167 countries was 22 places wide, and for ranks 68 to 100—where the published scores were most densely packed—it was 36 places wide. In a design with 131,072 model evaluations that also varied normalization and aggregation, the Sobol' indices showed that the Personal Freedom weight had the largest total effect (0.23). Furthermore, a weight's total effect was anticorrelated with the correlation ratio between that weight's pillar and the composite (Spearman -0.76). It is shown that a weight can only change country rankings through the share of its pillar's variance that is not shared with the composite, $1 - \rho_j^2$, multiplied by the squared ratio of that pillar's spread to the composite's spread, c_j^2 . This quantity is the rank leverage, $c_j^2(1 - \rho_j^2)$. Using arithmetic aggregation, rank leverage was found to follow the total effects of the weights under all four normalizations (Spearman 0.91 to 0.99). A pillar's alignment with the composite ordered the total effects of the weights closely only when the normalization gave the pillars equal spreads (Spearman correlation between the correlation ratio and the total effect of -0.39 on the published scores and -0.92 after z-scoring). The Personal Freedom and Social Capital pillars were nearly equally aligned with the composite but had different spreads. Their rank leverages differed by a factor of 3.72, and the total effects of their weights, with only the weights varied, differed by a factor of 3.61. Using equal nominal weights, rank leverages differed by a factor of six. Only the transformation from pillar scores to the index was perturbed, and thus the intervals do not represent uncertainty below the level of the pillars.

Keywords: composite indicators, Legatum Prosperity Index, uncertainty analysis, sensitivity analysis, Monte Carlo simulation, Sobol' indices, rank robustness, country rankings, weighting, correlation ratio, principal component analysis, leverage

1. Introduction

Composite indices transform hundreds of country indicators into a single value used in economic regressions, policy reports, and the news. Every composite index makes three decisions in its construction: how to normalize each indicator, how to weight each component, and how to aggregate the weighted components. Saisana et al. applied uncertainty and sensitivity analysis to these decisions in order to evaluate the reliability of national rankings (1). The handbook of the Organisation for Economic Co-operation and Development and the European Commission Joint Research Centre (OECD–JRC) also recommends that such an analysis be made as part of composite index construction (2).

Such an analysis is presented here for the 2023 Legatum Prosperity Index. The index has been studied in the fields of economics and policy (3), and the 2023 version reports scores for 167 countries from 2007 to 2023. The final step of the index is explicitly documented: the overall score is the equal-weighted arithmetic mean of twelve pillar scores (4). This property enables exact replication prior to perturbation. The methodology report for the index evaluates sensitivity to indicator weights and element weights (4). This study evaluates sensitivity to the pillar weights, normalization and aggregation, and whether the nominal weight of 1/12 for each pillar corresponds to its effective importance in the composite index, which in general does not have to be the case (5, 6).

Under three different weighting priors, the overall ranking remained stable, although individual mid-table positions did not. Under the uniform prior, the median 90 percent rank interval was 22 places wide. In the joint design, the weights on Personal Freedom and Natural Environment had the largest total effects. The weight with the largest effect on a country's rank varied by country. Across the twelve pillars, the total effect of each weight was anticorrelated with the correlation ratio between its pillar and the composite.

Section 6 explores this anticorrelation. It derives how each weight affects the composite and which part of that effect can change rankings. The variance that a change in a weight creates in the composite takes the form of an expression used by Schlossarek, Syrovátka and Vencálek (7) in the calculation of leave-one-out importance. This expression can be split into a rescaling term that changes no ranking and a rank leverage term, $c_j^2(1 - \rho_j^2)$. Here, ρ_j is the correlation between the pillar and the composite and c_j is the standard deviation of the pillar relative to that of the composite. The separation of this expression, the conditions under which alignment measures such as the correlation ratio order the influence of the weights, and the exact relationship between this separation and leave-one-out importance are the contribution of this study beyond the perturbation analysis. These conditions were tested on the 2023 index with sensitivity analyses that varied only the weights, under each of four normalizations. Section 8 separates uncertainty in the implementation of the index from uncertainty in its conceptual framework.

2. Related Work

Uncertainty and sensitivity analysis of composites. The methodological framework is from Saisana et al. (1): uncertainty in weighting, aggregation and normalization is propagated via Monte Carlo sampling, and the resulting distribution of ranks is decomposed using global sensitivity indices. The OECD–JRC handbook (2) describes the approach and the standard methods at each step. Greco et al. (8) reviewed the literature and pointed out that weighting cannot be performed independently of aggregation, because under compensatory aggregation weights represent trade-offs between criteria; with linear aggregation, equal weights imply the same trade-off across all units. In a review of 280 highly cited studies from nineteen different domains, 41 percent employed global sensitivity methods and 34 percent employed one-factor-at-a-time methods, even though more than half employed clearly non-linear models (9).

Variance-based sensitivity analysis. The variance-based sensitivity measures used here are from Sobol' (10). Designs and estimators that obtain first-order and total-effect sensitivity measures in $N(k + 2)$ model evaluations (11) are implemented in SALib (12), which was used here for the sampling design. Diebold (13) estimated first-order and total-effect Sobol' indices for the Elite Quality Index and found that its 2021 ranking was relatively stable for the top 50 countries but considerably uncertain for middle and lower ranking countries.

Weights versus importance. Paruolo et al. showed that nominal weights are not a measure of the importance of variables (5) and proposed each variable's main-effect correlation ratio η^2 with the composite as a measure of importance in their place. Becker et al. (6) proposed methods to close this gap between weights and importance. Schlossarek, Syrovátka and Vencálek (7) approached the problem from the opposite direction, defining the most important variable as the one whose removal from the composite changes the results most, and deriving the correlation between an index and the index computed without that variable. Their measure is related in Section 6 to the weight-perturbation sensitivity analyzed here. Kuc-Czarnecka (14) used the variance-based method on the Human Development Index for 189 countries and found main-effect correlation ratios of 0.537 for income, 0.388 for health and 0.208 for education, compared with nominal weights of one third for each variable. Weights optimized toward equal normalized correlation ratios changed ranks by 1.68 places on average.

Data-driven weighting and latent structure. Mazziotta and Pareto pointed out that principal component analysis implicitly assumes a reflective measurement model, which is not appropriate for all indicators (15). Across 68 specifications of an Italian well-being index, Cantone and Tomaselli found that principal component weighting increased specification uncertainty, whereas the choice of aggregation method did not have a large effect on it (16). In this study, the pillars were considered formative, meaning that they make up the construct rather than reflect it. A pillar that showed only a weak correlation with the other pillars was therefore not considered less relevant, although it would be given less weight under first-principal-component weighting. For these reasons, Section 4 used principal component analysis descriptively, to characterize the correlation structure and effective dimensionality of the data, and the composite based on the first principal component in Section 8 is presented as one possible construction but not as a preferred weighting method.

Previous research related to the Legatum index and to publishing uncertainty. Budsaratragoon and Jitmaneeroj (3) utilized cluster analysis, data mining, importance–performance analysis and partial least squares path modeling in analyzing the index for 142 countries. They found that the index dimensions should not be weighted equally when formulating policy. Their focus was on the prioritization of policy, not the robustness of the index ranking. Matsuda dos Santos and Peterson (17) evaluated the reliability of the data used in the index and used cluster analysis to group countries into configurations. The 2023 methodology report for the index evaluated the sensitivity of the 2019 rankings to changes in element and indicator weights (4). However, no published study was found that evaluated the sensitivity of the 2023 ranking to aggregation, normalization or pillar weights. Kuc-Czarnecka, Lo Piano and Saltelli (18) illustrated how stakeholders who accept the same data but have different ideas about what is relevant can create composite indices that yield different results. They also argued that the creators of an index should be the first to evaluate its robustness. This study evaluates the sensitivity of the ranking to the index constructions that were not selected.

3. Data and Baseline Replication

The 2023 Legatum Prosperity Index uses a three-level structure to compute a score for 167 countries. Three hundred indicators are normalized using the distance-to-frontier method and aggregated into 67 elements. These elements are then aggregated into the twelve pillars of the index, each scored from 0 to 100. The twelve pillars are Safety and Security, Personal Freedom, Governance, Social Capital, Investment Environment, Enterprise Conditions, Infrastructure and Market Access, Economic Quality, Living Conditions, Health, Education, and Natural Environment. The scores for each pillar are then combined using an equal-weighted arithmetic mean to produce the overall index score for each country (4). The published table of pillar scores was used for this analysis. Only the transformation from pillar scores to overall index score was perturbed. The choice of indicators, imputation of missing values, normalization of the raw indicators using the distance-to-frontier method, and all aggregation below the level of the pillars remained as in the published index. The rank intervals reported below vary only the weights applied when combining the pillar scores into an overall index score, with normalization and aggregation at this step held fixed. They are therefore not estimates of the overall methodological uncertainty of the index. This scoping decision is discussed in Section 10.

Given a weight vector w with twelve non-negative entries that sum to one, the composite score for country i is $Y_i = \sum_j w_j s_{ij}$, where s_{ij} is the score of country i on pillar j , and the rank of country i is one plus the number of countries with higher composite scores. The published index corresponds to the weights $w = (1/12, \dots, 1/12)$.

Using the published data file of pillar scores, the equal-weighted arithmetic mean had reproduced the overall index score for each country with a maximum absolute error of 5.0×10^{-11} and had reproduced the rank of every country (Spearman and Kendall correlations of exactly 1.0 across all 167 countries) before any perturbation was applied. Thus, any reported rank interval is free from the reconstruction error that would otherwise confound the perturbation analysis.

4. Latent structure of the twelve pillars

Prior to any weights being perturbed, the structure of the pillars was examined. What a perturbation can change depends on how much the pillars already share. As this section appears prior to the Materials and Methods section, the methods used are described inline. All twelve pillars are positively correlated with each other. The correlation matrix has a condition number of 265.2. Variance inflation factors are greater than fifteen for Infrastructure and Market Access (19.5), Investment Environment (19.1), Governance (15.9) and Living Conditions (15.1), but are only 2.5 for Social Capital and Natural Environment. Therefore, the twelve pillars are not twelve independent measures.

A principal component analysis of the correlation matrix found a first component that accounted for 74.5 percent of the total variance. All pillars had positive loadings (correlations between each pillar and the component) between 0.69 (Natural Environment) and 0.95 (Investment Environment). Therefore, the first component is a dominant common-variation component and not a contrast. No substantive label is assigned to this component here. A second component accounted for a further 8.9 percent of the total variance and separated Health (0.39), Living Conditions (0.34) and Education (0.29) from Personal Freedom (−0.54) and Natural Environment (−0.51). No other component accounted for more than 4.5 percent of the total variance.

Using the Kaiser criterion, two components were retained. The second eigenvalue was 1.071. Using Horn's parallel analysis method (19) as modified by Glorfeld (20), one component was retained. In 1,000 matrices of the same size filled with independent draws from a standard normal distribution, the 95th percentile of the second eigenvalue was 1.424. Since the 167 countries in the dataset are considered to be a population and not a sample, this value was used as a criterion for eigenvalues that would occur by chance and not as a test. Both criteria are shown in Figure 1. These criteria differ only on the second component. If all pillars shared a single dimension perfectly, then all weight vectors w would assign the same ranking to all countries. The remaining 25.5 percent of the variance lies outside the first component, and reweighting can reorder countries only through that part.

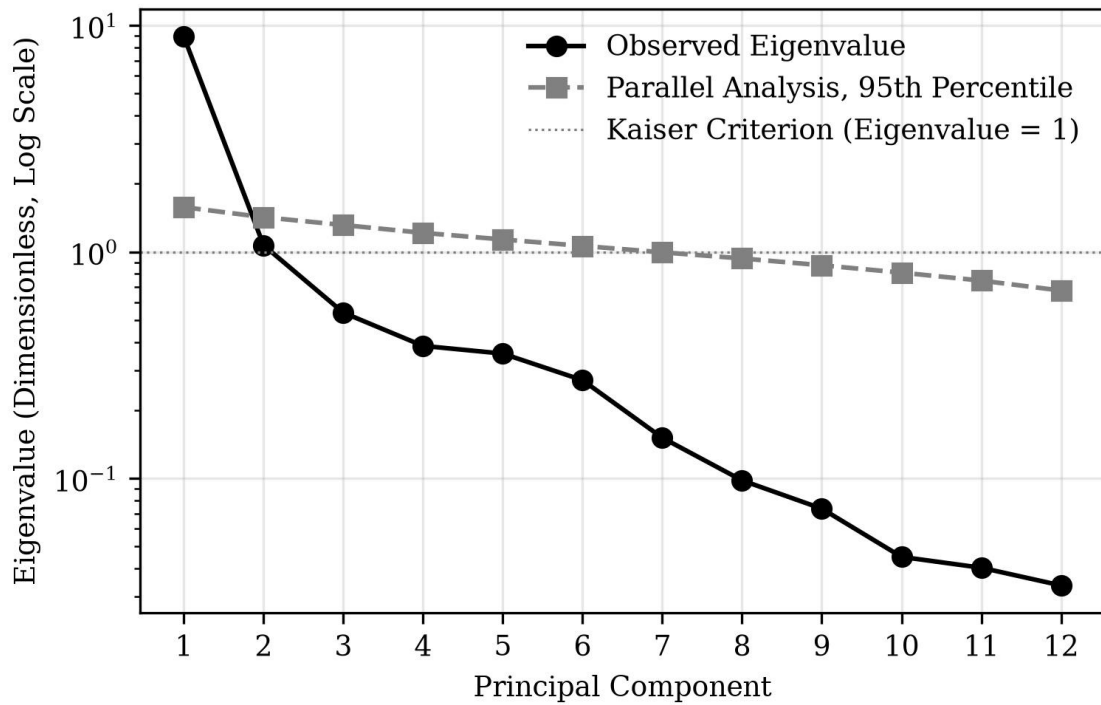


Figure 1. Eigenvalues of the correlation matrix of the twelve pillar scores of the 167 countries, in decreasing order, against the parallel-analysis threshold (95th percentile of each ordered eigenvalue over 1,000 random standard-normal data sets of 167×12) and the Kaiser criterion (eigenvalue of one). Only the first eigenvalue exceeds the parallel-analysis threshold. Logarithmic vertical scale.

The correlations shown above are marginal. Two pillars might be correlated only because both capture the same dominant dimension. Therefore, a Gaussian graphical model was fit. Each edge was the partial correlation between two pillars controlling for the remaining ten. No shrinkage was performed, given that there are 167 observations and only twelve variables, so the problem is not high-dimensional. Furthermore, the graphical lasso (21) is unreliable for model selection at this ratio of observations to variables (22). Instead, each partial correlation was transformed to a two-sided p-value using the normal-theory Fisher z transform, with a standard error of $1/\sqrt{(167 - 10 - 3)}$. Each of the sixty-six partial correlations was set to zero unless it survived the Benjamini and Hochberg procedure (23) at $q = 0.05$. As with parallel analysis, this was used as a criterion for removing small partial correlations, but not as a test. Sixteen edges were retained. Table 1 shows the strength and expected influence (24) of each pillar. Expected influence is shown because strength disregards the sign of the edges: positive and negative edges that offset each other both add to strength but cancel in expected influence. The unthresholded partial correlations can be found in the repository. Betweenness and closeness were not calculated because both assume flow between nodes along the shortest paths, which is not captured by a graph of partial correlations (25).

Table 1. Strength and expected influence in the partial correlation network of the twelve pillars, with edges selected by the Benjamini and Hochberg procedure at $q = 0.05$. Strength is the sum of absolute surviving partial correlations incident on a pillar; expected influence is the signed sum; edges is the number of the eleven possible edges that survive.

Pillar	Strength	Expected influence	Edges
Living Conditions	1.99	0.92	6
Personal Freedom	1.75	0.79	5
Enterprise Conditions	1.42	0.87	4
Governance	1.32	1.32	3
Investment Environment	1.00	1.00	3
Infrastructure and Market Access	1.00	1.00	3
Natural Environment	0.69	0.17	2
Health	0.59	0.11	2
Education	0.47	0.47	1
Economic Quality	0.46	-0.01	2
Social Capital	0.22	0.22	1
Safety and Security	0.00	0.00	0

Living Conditions and Personal Freedom had the highest strength, with six and five surviving edges, respectively, while Safety and Security had no edges. However, at $q = 0.10$, every pillar had at least one edge. Strength is not the same as predictability. Social Capital, Natural Environment and Safety and Security have the lowest variance inflation factors (2.5 to 2.8) and are the least predictable given the other eleven pillars. The strengths of these three pillars are 0.22, 0.69 and zero, respectively. In contrast, Personal Freedom is second in strength and has the fourth-lowest variance inflation factor (5.2). Strength and the loadings on the first principal component are compared to realized importance, leverage and the total-effect indices in Section 7.6.

5. Materials and Methods

5.1 Weight-uncertainty propagation

The first exercise kept the published pillar scores and the arithmetic mean and varied only the weights, drawing $M = 10,000$ weight vectors from each of three priors (Appendix A.2).

The uniform prior is the flat Dirichlet distribution over all twelve-component weight vectors that are non-negative and sum to 1, so any pillar may carry nearly all of the weight or nearly none.

The near-equal prior is a concentrated Dirichlet distribution (concentration $c = 50$) that keeps each weight close to $1/12$, with a standard deviation of about 0.011, and represents an analyst who accepts equal weighting but not its exactness.

The grid prior assigns each weight independently one of the values $\{0.5, 1, 1.5, 2\}$ and rescales the vector to sum to 1, so no weight exceeds four times another. These are the values the index producer typically assigns to indicators within an element and drew at random in its own sensitivity test of indicator weights (4); here they were applied to the pillar weights.

5.2 Joint perturbation design

The second exercise allowed all three construction choices to be varied simultaneously. The Sobol' sampling design requires that the model inputs form a unit hypercube, so fourteen coordinates in $[0,1]$ were used: twelve for the pillar weights and two additional coordinates that determine the normalization and aggregation methods. The twelve weight coordinates were transformed into a weight vector whose components sum to 1 through a standard map that reproduces the uniform prior of Section 5.1 exactly (Appendix A.3). The remaining two coordinates each selected one option from a discrete menu by dividing $[0, 1]$ into equal intervals, one per option.

Normalizations (four options): the native distance-to-frontier scale used by Legatum; min–max normalization to the $[0,100]$ range; z-scoring (standard scoring); within-pillar percentile rank.

Aggregators (two options): weighted arithmetic mean (fully compensatory: a surplus in one pillar can be balanced out by a deficiency in another); weighted geometric mean (partially compensatory: a surplus in one pillar is only partly balanced by a deficiency in another, and every pillar score must be positive; see Appendix B).

These are the standard alternatives from the composite-indicator literature (2). For each of the 167 countries, the model output was its rank $r_i(x)$ under the perturbed choices.

5.3 Measures

Rank shift and intervals. For each prior, the average absolute rank shift $\bar{R}_S$ of Saisana et al. (1) was computed over all countries and draws (Appendix A.1). For each country, the median simulated rank, the interquartile range and the 90 percent rank interval (5th to 95th percentile of the simulated ranks) were computed from the 10,000 draws of a prior, the uniform prior unless stated otherwise; the width of the 90 percent interval is the measure of rank uncertainty. Spearman and Kendall correlations with the published ranking were computed on every 50th draw (200 draws per prior).

Sobol' indices. For each of the fourteen inputs, the Sobol' indices measure the contribution of that input (and its interactions with other inputs) to the total variance in the rank of a country. Two indices are provided for each input: the first-order index S_j measures the fraction of the variance in the country's rank that is due to input j alone; the total-effect index S_{Tj} measures the fraction of the variance in the country's rank that is due to input j , either alone or in interaction with any of the other inputs.

Definitions are given in Appendix A.4. The Saltelli design, built from a scrambled Sobol' sequence, was generated with SALib (12) at a base sample size of $N = 8192$, giving $N(k' + 2) = 131,072$ model evaluations for the $k' = 14$ inputs, and at $N = 1024$ as a check (Section 7.4). For each country, first-order indices were computed with the estimator of Saltelli et al. (11) and total-effect indices with the Jansen estimator given there, both implemented directly in the analysis

code. Ninety-five percent intervals were obtained by resampling the rows of the base sample 200 times with replacement, so they measure Monte Carlo estimation error only. Because rank variance differs across countries (Denmark's rank barely moves, whereas mid-table ranks move considerably), the per-country indices were pooled with weights proportional to each country's rank variance, which makes each pooled index the share of the rank variance summed over all countries that is attributable to that input (Appendix A.4). Any average hides differences between countries, so the per-country indices were also analyzed, in Sections 7.7 to 7.9.

Realized importance. Following Paruolo et al. (5), the realized importance of pillar j is its correlation ratio η^2_j , the fraction of the variance of the composite explained by that pillar's scores alone (Appendix A.5); each pillar's share of the summed η^2 was compared with its nominal share of 1/12. Because the composite contains pillar j , η^2_j does not measure how far the pillar can be predicted from the other eleven, which the variance inflation factors of Section 4 measure. Appendix Table A1 also reports η^2_{-j} , computed against the equal-weighted mean of the other eleven pillars.

Weights-only designs. To match the setting of Section 6, in which the pillar scores and arithmetic aggregation are fixed, a separate Saltelli design was run for each of the four normalizations with arithmetic aggregation, sampling only the twelve weight coordinates ($N = 8192, 114,688$ evaluations each). Total effects were estimated and pooled as above, with 200 bootstrap resamples on the published scores and 100 under each other normalization. The first-component loadings and network centralities of Section 4, η^2_j , the leverage quantities of Section 6 and the total effects of both designs were compared across the twelve pillars by Spearman and Pearson correlation.

Computation. The analysis used Python 3.12.10 with NumPy 2.5.1, SciPy 1.18.0, pandas 3.0.5, scikit-learn 1.9.0 and SALib 1.5.2, and the figures were drawn with Matplotlib 3.11.1. All reported results used random seed 42.

5.4 Assumptions

The analysis rests on the following assumptions, each stated here with its justification and its bearing on the results.

(i) *Sobol' indices are reported with respect to the hypercube coordinates, not the weights.*

Variance-based indices require independent inputs, and the twelve weights are not independent: they lie on the simplex and sum to one. The model inputs are therefore the twelve independent uniform coordinates of the Saltelli design, carried onto the simplex by the map of Appendix A.3. A reference to the total effect of a pillar's weight is shorthand for the total effect of the coordinate generating it, whose variation necessarily rescales the other eleven. The map was used because the weights it induces follow the flat Dirichlet prior exactly, but another map onto the same prior could divide the rank variance among the coordinates differently.

(ii) *The menus of normalization and aggregation alternatives are treated as equally plausible.*

The normalization selector was drawn uniformly over four alternatives and the aggregation

selector over two, so the total effect attributed to either is conditional on the menu in Appendix B; a fifth normalization would change it. The standing of the two selectors among the pillar weights depends on these alternatives and on the positivity shift of (vii), and is not a general ordering.

(iii) *Pillar scores are treated as fixed and free of measurement error.* No uncertainty in the 300 underlying indicators, their normalization, weighting and imputation, or their aggregation into 67 elements and twelve pillars was propagated, so the intervals of Section 7.3 omit that uncertainty (Section 10).

(iv) *The three priors treat the twelve pillars symmetrically.* Each prior is exchangeable across pillars, asserting that no prior knowledge favors one over another. That is a normative position, not a neutral one, and a reader with substantive views about which pillars matter should read the intervals as describing uncertainty around equal weighting, on which all three priors are centered, rather than around their own preferred weights.

(v) *The flat Dirichlet prior is agnostic in a specific sense.* It is uniform over the space of valid weight vectors, but not invariant under reparameterization, and not the only distribution that could be called non-informative. It gave the widest rank intervals of the three priors, a median 90 percent width of 22 places against 11 under the grid prior and 4 under the near-equal prior, and it underlies the intervals of Section 7.3, every Sobol' design and Sections 7.7 to 7.9, which are therefore conditional on the most diffuse prior considered.

(vi) *Rank is treated as a square-integrable model output.* Rank is bounded and discrete, which suffices for the functional decomposition, but a variance share of such an output describes rank dispersion rather than decomposing a continuous physical quantity.

(vii) *The positivity shift required by geometric aggregation is one repair among several.* The shift of Appendix B was applied to every column with non-positive entries: all min–max columns, whose minimum is zero, and all z-score columns. It set each such column minimum to 10^{-6} , under which the geometric min–max and z-score composites ranked countries almost identically (mean absolute rank difference 0.03 places over 1,000 uniform-prior draws).

(viii) *The correlation ratio was estimated by quantile binning.* Equal-count bins were used so that each bin mean rested on sixteen or seventeen countries at the ten bins used in the text. Estimates at five, ten and twenty bins are reported in Appendix Table A1. Levels move by as much as 0.136 across that range while the ordering is close to stable, the rank correlation between the five-bin and twenty-bin estimates being 0.972.

(ix) *The 167 countries constitute the population of the index rather than a sample from it.* No inference to a wider population was drawn: correlations across pillars and countries are descriptive, the Fisher z p-values and false-discovery-rate threshold of Section 4 served only as a rule for discarding small partial correlations, and the intervals described in Appendix A.5 and the resampling check of Section 7.9 are stability diagnostics.

(x) *The rank intervals are marginal, not simultaneous.* Coverage holds for each country separately, not for the ranking as a whole (Section 7.3).

(xi) *The analysis is a single cross-section of the 2023 edition.* The stability of the sensitivity structure across editions was not examined. The 2026 edition has since restructured the index, which bears on the scope of the conclusions and is discussed in Section 8.

(xii) *Per-country Sobol' indices were pooled with weights proportional to rank variance (eq10).* The pooled indices therefore describe rank variance summed over countries (Section 5.3), half of which falls on 33 countries, 31 of them ranked 34 to 134. Under an unweighted mean the Personal Freedom and Natural Environment weights still led, but the aggregation selector exceeded five pillar weights rather than eight and the normalization selector seven rather than four (Section 7.4). The per-country indices of Sections 7.7 to 7.9 do not depend on this choice.

(xiii) *Leverage is a local, score-level quantity.* The identities (eq4) and (eq5) describe score changes at equal weights under arithmetic aggregation of a fixed normalized matrix, whereas total effects describe rank variance across the whole simplex. Their agreement is therefore an empirical result, tested in Section 6 against weights-only Sobol' designs with arithmetic aggregation and each normalization held fixed; the derivation does not cover geometric aggregation or the joint design of Section 7.4.

(xiv) *Dependence among pillars was treated as linear, and sensitivity profiles as points in Euclidean space.* Linear measures match the additive construction of the index but capture linear association only; this applies to the principal components and partial correlations of Section 4 and to the regressions of Section 7.7. The Fisher z rule of Section 4 also treats the pillar scores as multivariate normal, and the retained edges depend on its threshold. Ward clustering (Section 7.9), whose criterion assumes Euclidean distance, was applied to shares that sum to one; on centered log-ratio shares it gave a largely different six-cluster partition (adjusted Rand index 0.08).

6. Leverage and rank leverage

6.1 *The perturbation induced by the sampling design*

The map of Appendix A.3 sets $w_j = t_j / T$, with $t_j = -\ln u_j$ and $T = \sum_1 t_i$. Differentiating with respect to one coordinate gives

$$\partial w / \partial t_j = (e_j - w) / T \quad (\text{eq1})$$

so a change in t_j moves the weight vector along the simplex direction $(e_j - w)$, where e_j is the j th unit vector: the weight on pillar j rises and the other eleven contract in proportion. The same direction arises whenever independently drawn positive weights are renormalized; the map of normalized exponential variates is the case in which the weights follow the flat Dirichlet distribution exactly. For a step of size h along $(e_j - w)$, the composite score of country i changes by

$$\Delta Y_i = h (s_{ij} - Y_i) \quad (\text{eq2})$$

exactly, because the composite is linear in the weights (Appendix D.2).

6.2 *The rank-changing part of leverage*

The cross-country variance of this change, relative to the variance of the composite and per unit h^2 , is the leverage of the weight:

$$L_j = \text{Var}_i (s_{ij} - Y_i) / \text{Var}_i (Y_i) \quad (\text{eq3})$$

With $c_j = \sigma_j / \sigma_Y$, the ratio of the pillar's standard deviation to the composite's, and ρ_j their correlation,

$$L_j = (c_j - \rho_j)^2 + (1 - \rho_j^2) \quad (\text{eq4})$$

Leverage describes composite scores, not ranks. Regressing s_j on Y across countries, $s_j = m_j + a_j Y + \varepsilon_j$ with slope $a_j = \rho_j c_j$, splits (eq4) into

$$L_j = (a_j - 1)^2 + c_j^2 (1 - \rho_j^2) \quad (\text{eq5})$$

After a step along $(e_j - w)$ the composite is $(1 + h(a_j - 1))Y + hm_j + h\varepsilon_j$. The first two terms rescale and shift every country's score in the same way and change no rank, so only the second term of (eq5), the variance of ε_j relative to that of the composite, can reorder countries (Appendix D.5). That term, $R_j = c_j^2(1 - \rho_j^2)$, is called the rank leverage of the weight. It is the product of the share of the pillar's variance not shared with the composite, $1 - \rho_j^2$, and the squared relative spread, c_j^2 . A pillar perfectly aligned with the composite moves no rank (given $0 < a_j < k$, Appendix D.5), whereas a closely aligned pillar with a large spread can. Leave-one-out importance (7) depends on the same two quantities: the correlation between the composite and the equal-weighted mean of the other $k - 1$ pillars equals $f_j / \sqrt{(f_j^2 + R_j/(k - 1)^2)}$, with $f_j = (k - a_j)/(k - 1)$ and $k = 12$ (Appendix D.5).

6.3 *When alignment should order weight sensitivity*

Equation (eq5) states when a measure of alignment alone should order the rank leverage of the weights. If all pillars have the same spread, as after z-scoring, c_j is constant and R_j falls strictly as ρ_j^2 rises; the correlation ratio η^2 and first-component loadings, which track ρ_j^2 closely but not exactly, should then order it approximately. If every pillar contributes equally to the variance of the composite, then $a_j = 1$ for every j , since the mean of a_j is exactly one (eq17) (26), so that $c_j = 1/\rho_j$ and $R_j = (1 - \rho_j^2)/\rho_j^2$, which also falls strictly as ρ_j^2 rises. When spreads differ, as for the published pillar scores, whose standard deviations span a factor of 2.20, the squared relative spread can outweigh alignment. These are statements about rank leverage at a given set of weights; whether they carry over to global, rank-based Sobol' indices was tested empirically.

6.4 Tests on the 2023 index

Rank leverage was compared with the total-effect indices of the twelve weights in the weights-only designs of Section 5.3, one for each normalization under arithmetic aggregation, with R_j computed on the correspondingly normalized scores. Rank leverage tracked the total effects in every case (for the published scores, see Section 7.6), at Spearman correlations of 0.91 on the published scores, 0.94 after min–max normalization, 0.99 after z-scoring and 0.99 after percentile ranking. On the published scores the Pearson correlation was 0.99, and 0.98 with Personal Freedom, the largest value, excluded. Leverage (eq4) correlated with the total effects at Pearson 0.92, and 0.83 without Personal Freedom, and ordered the pillars as well as rank leverage did: each ordered 58 of the 66 pillar pairs correctly, against 44 for ρ^2 and 42 for η^2 . A step of $h = 0.05$ along $(e_j - w)$ from equal weights moved ranks by mean absolute shifts that correlated with R_j at Pearson 0.97 (Spearman 0.80).

The results were consistent with the conditions of Section 6.3. The Spearman correlation between a weight's total effect and its pillar's η^2 was -0.39 on the published scores, -0.62 after min–max normalization, -0.92 after z-scoring and -0.98 after percentile ranking, so alignment ordered weight sensitivity closely only where normalization made the pillar spreads comparable. The anticorrelation of -0.76 in the joint design (Section 7.5), which also varies aggregation, pools these settings.

Personal Freedom and Social Capital isolate the spread factor. Their squared correlations with the composite were 0.5823 and 0.5846, and their standard deviations 19.91 and 10.35, against 13.31 for the composite. The rank leverage of the Personal Freedom weight was 3.72 times that of the Social Capital weight; with only the weights varied on the published scores, the ratio of their total effects was 3.61 (95 percent bootstrap interval 3.42 to 3.80). The ratio of their leverages (eq4) was 2.30, because leverage also counts the rank-neutral rescaling, which made up 0.164 of Social Capital's leverage of 0.416 and 0.020 of Personal Freedom's 0.955. The correlation ratio ordered the pair the wrong way (0.676 against 0.612). For the next three pairs closest in ρ^2 , the ratios of rank leverage and of total effects were 0.49 and 0.58 (Economic Quality to Education), 1.01 and 1.01 (Investment Environment to Infrastructure and Market Access), and 0.46 and 0.59 (Enterprise Conditions to Living Conditions). On the 2026 variant with uniform weights and arithmetic means at every node (161 countries), whose index is the mean of its three domains, (eq4) and (eq17) held to 5.6×10^{-16} , a check that the algebra had been implemented correctly.

7. Results

7.1 Replication is exact

The replication of Section 3 was exact, so the pillar-to-index step contains no undocumented processing.

7.2 The ranking as a whole is stable under all three priors

Table 2 and Figure 2 report the weight-uncertainty results. The average absolute rank shift $\bar{R}_S$ rose from 0.98 places under the near-equal prior to 2.93 under the grid prior and 5.68 under the uniform prior, which allows any single weight to approach zero or one. Under the uniform prior the median Spearman correlation with the published ranking was 0.9899 and its 5th percentile 0.9658, computed on 200 of the 10,000 draws. This stability of the ordering as a whole is consistent with the one dominant dimension shared by the twelve pillars (Section 4).

Table 2. Weight-uncertainty analysis of the 167 countries, 10,000 weight draws per prior. $\bar{R}_S$ is the average absolute rank shift from the published 2023 ranking, in places; Spearman and Kendall are the rank correlations with the published ranking, computed on every 50th draw (200 draws per prior). Lower $\bar{R}_S$ and higher correlations indicate a more stable ranking.

Weighting prior	$\bar{R}_S$	Median Spearman	5th percentile Spearman	Median Kendall	5th percentile Kendall
Near-equal prior	0.98	0.9996	0.9989	0.988	0.980
Grid prior {0.5, 1, 1.5, 2}	2.93	0.9965	0.9911	0.956	0.930
Uniform prior (agnostic)	5.68	0.9899	0.9658	0.922	0.850

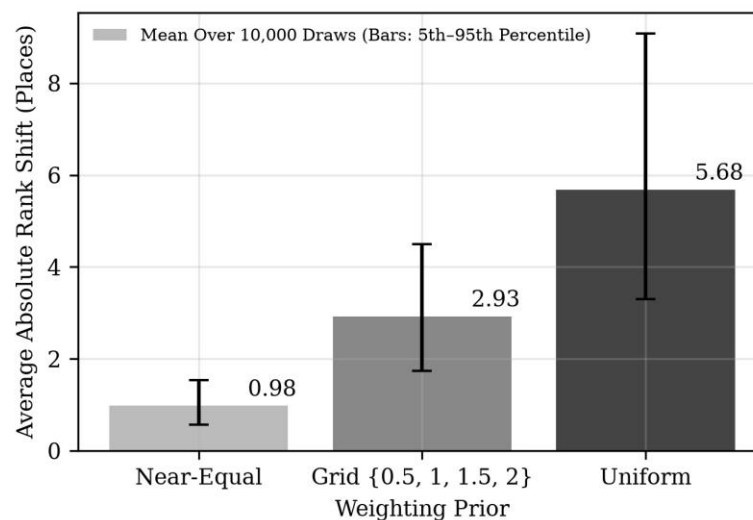


Figure 2. Average absolute rank shift of the 167 countries from the published 2023 ranking under the near-equal, grid and uniform weighting priors. Bars give the mean over 10,000 weight draws per prior, and error bars the 5th to 95th percentile across those draws.

7.3 Individual mid-table positions are not stable

Figure 3 plots the 90 percent rank intervals of the 167 countries under the uniform prior. Their median width was 22 places, against 11 under the grid prior and 4 under the near-equal prior. At the two ends of the table, Denmark (published rank 1) had an interval of [1, 3] and South Sudan (rank 167) one of [165, 167]. Intervals were wider in the middle of the table: Turkmenistan (rank 107) had [82, 137], Belarus (rank 78) [53, 105] and China (rank 54) [44, 92]. No published rank fell outside its own interval. Appendix Table A2 lists the widest and narrowest intervals, and `outputs/rank_uncertainty.csv` in the repository gives all 167.

These intervals are marginal: each attains its coverage for one country considered alone, and together the 167 intervals are not a 90 percent statement about the whole ranking (27).

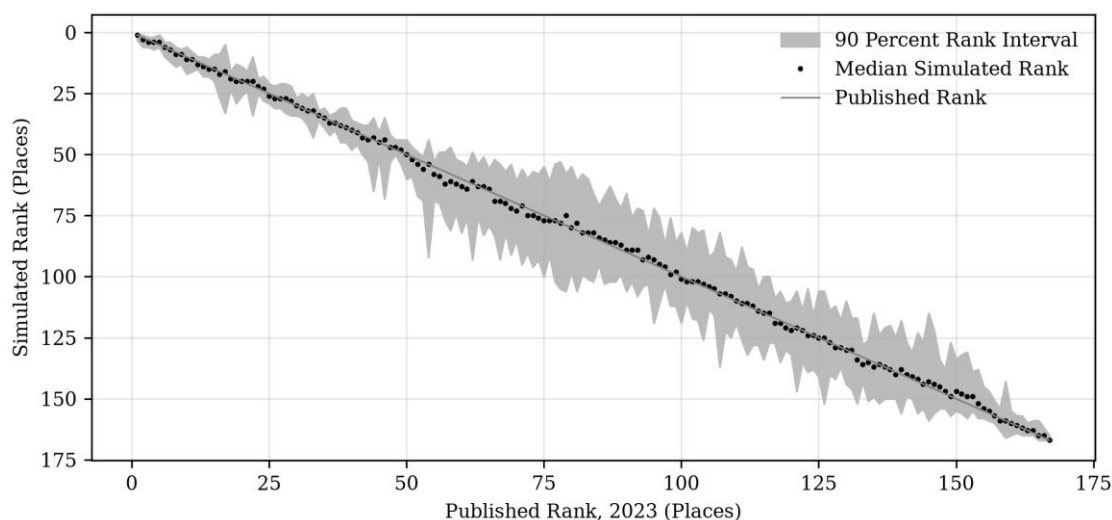


Figure 3. Simulated rank of each of the 167 countries under the uniform weighting prior (10,000 weight draws) against its published 2023 rank. The 90 percent rank interval runs from the 5th to the 95th percentile of a country's simulated ranks.

Figure 4 plots interval width against published rank, and Table 3 summarizes width by tier. Median width peaked at 36 places for published ranks 68 to 100, and the same tier had the widest median intervals under the grid prior (19 places) and the near-equal prior (6 places). Section 7.8 examines what makes an interval wide.

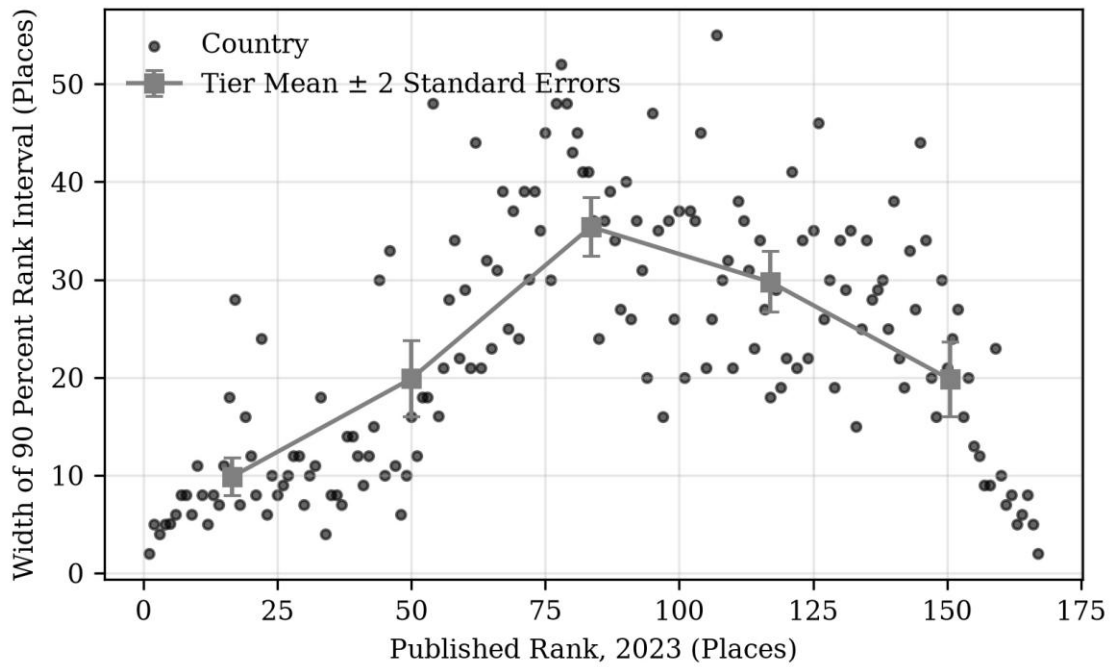


Figure 4. Width of the 90 percent rank interval (5th to 95th percentile of simulated ranks, 10,000 uniform-prior weight draws) of each of the 167 countries against its published 2023 rank, with the mean width of each of the five tiers of Table 3 (ranks 1–33, 34–67, 68–100, 101–134 and 135–167) and error bars of two standard errors of that mean.

Table 3. Width, in places, of the 90 percent rank interval (5th to 95th percentile of simulated ranks) by tier of the published 2023 ranking, uniform prior, 10,000 weight draws. Each tier holds 33 or 34 countries.

Published rank tier	Mean width	Median width	Max width
1–33	9.8	8.0	28
34–67	19.9	17.0	48
68–100	35.4	36.0	52
101–134	29.8	29.5	55
135–167	19.8	20.0	44

7.4 Sobol' indices of the joint design

Figure 5 and Appendix Table A3 give the variance-weighted Sobol' indices of the fourteen inputs of the joint design (Section 5.2), with 95 percent bootstrap half-widths. The Personal Freedom weight had the largest total effect, $S_T = 0.231 \pm 0.007$ ($S = 0.191 \pm 0.009$), followed by the Natural Environment weight at $S_T = 0.142 \pm 0.005$ ($S = 0.107 \pm 0.006$); the main effects of these two weights alone accounted for 0.298 of the rank variance. The aggregation selector reached $S_T = 0.086 \pm 0.003$, exceeding eight of the twelve pillar weights, and the normalization selector $S_T = 0.067 \pm 0.002$, exceeding four. Both counts depend on the positivity shift of Appendix B: with each shifted column's minimum set to 1 or 5 percent of its range instead of 10^{-6} , the total effect of the aggregation selector fell to 0.074 and 0.065, exceeding six and three pillar weights, and that of the normalization selector fell to 0.059 and 0.057, exceeding three and

one, while no pillar-weight total effect changed by more than 0.004. The smallest total effect was 0.055 ± 0.002 , for the Investment Environment weight.

The first-order indices of the fourteen inputs summed to 0.867, so most of the rank variance came from main effects. Interactions, measured as $1 - S/S_T$, made up 17 to 42 percent of the total effect of each pillar weight, 45 percent of that of the aggregation selector and 66 percent of that of the normalization selector ($S = 0.023$ against $S_T = 0.067$), whose first-order index alone therefore understated its total effect by a factor of 2.9.

Appendix Table A3 also reports a run at $N = 1024$. Averaged across the twelve pillar weights, bootstrap half-widths at $N = 8192$ were smaller by a factor of 2.92 for the total-effect indices and 2.97 for the first-order indices, close to the factor of 2.83 expected from an eightfold increase. The largest change in any index was 0.007, and the ordering of the fourteen inputs by total effect was unchanged except that the normalization selector moved from eighth place at $N = 1024$ to tenth at $N = 8192$, exchanging places with the Living Conditions and Enterprise Conditions weights, whose intervals overlap its own at both sample sizes.

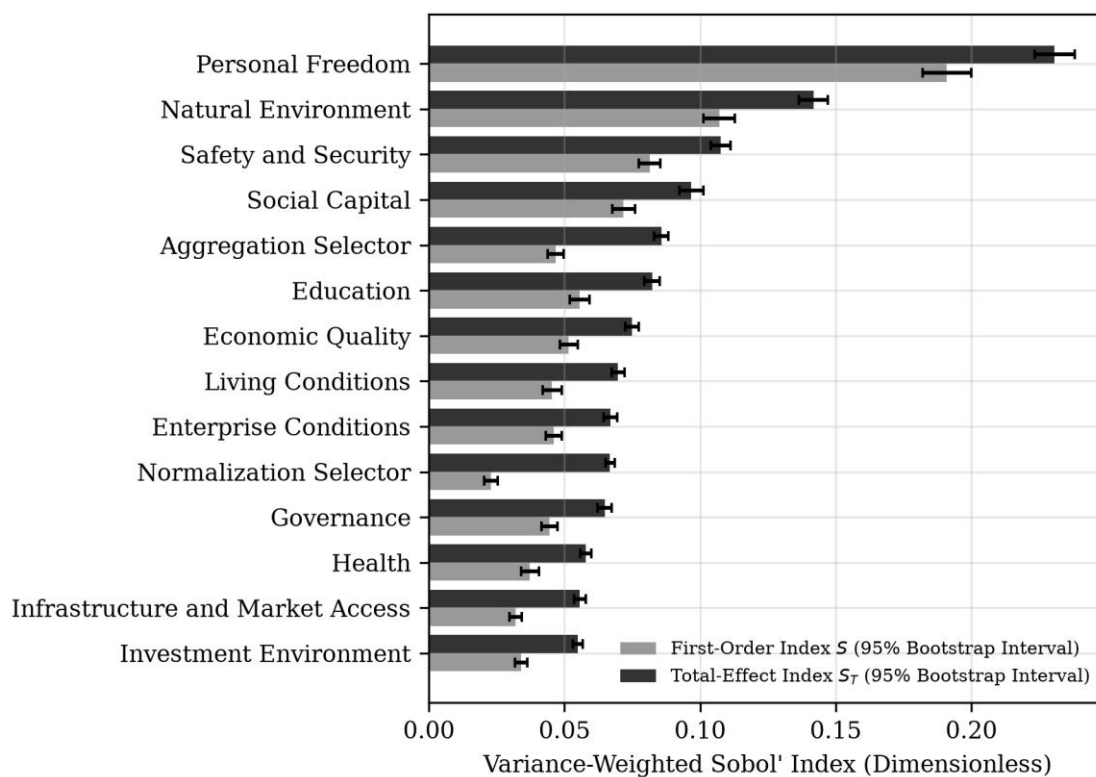


Figure 5. First-order (S) and total-effect (S_T) Sobol' indices of country rank for the fourteen inputs of the joint design (twelve pillar weights under the uniform prior, and the normalization and aggregation selectors), averaged over the 167 countries with weights proportional to each country's rank variance. Saltelli design with the first-order estimator of Saltelli et al. and the Jansen total-effect estimator, base sample $N = 8192$ (131,072 model evaluations); error bars are 95 percent bootstrap intervals from 200 resamples.

7.5 Realized importance

Figure 6 plots the nominal and realized importance of the twelve pillars (full table in Appendix Table A1). Realized importance shares ran from 0.095 for Infrastructure and Market Access ($\eta^2 = 0.90$) to 0.060 for Natural Environment ($\eta^2 = 0.56$), against a nominal share of 0.083; Personal Freedom ($\eta^2 = 0.68$) and Social Capital ($\eta^2 = 0.61$) joined Natural Environment at the low end.

The exclusion-based estimate η^2_{-j} , against the mean of the other eleven pillars, lowered every value by 0.02 to 0.07, most for Personal Freedom (0.067), and left the ordering unchanged (Spearman 1.000). A high η^2 does not make a pillar dispensable: in a formative index, removing a pillar changes the construct (15), and the five pillars with the highest η^2 loaded at 0.89 to 0.95 on the first principal component, which marks them as central to the common dimension.

Across the twelve pillars, the total effect of a weight in the joint design was anticorrelated with its pillar's η^2 (Spearman -0.76 ; -0.80 for the first-order indices). This is an empirical result for this edition and design, and Section 6.4 shows that it depends on the normalization. The four largest total effects belonged to the four pillars least predictable from the other eleven, those with the lowest variance inflation factors (2.47 to 5.16, against at least 6.53 for the other eight).

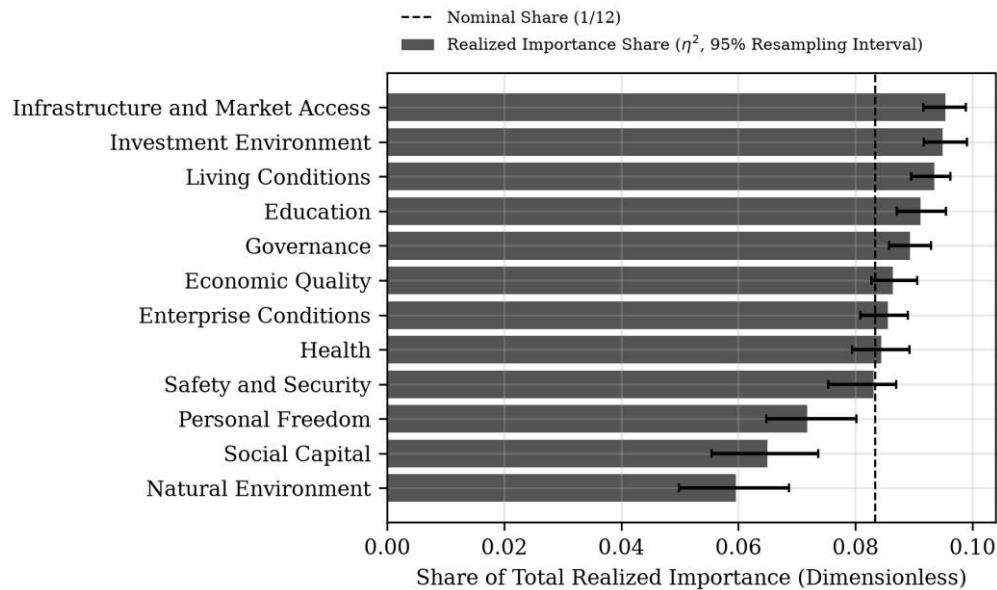


Figure 6. Realized importance (share of main-effect η^2) per pillar against the uniform nominal weight 1/12. Error bars are 95 percent resampling intervals over the 167 countries, used as stability diagnostics (Appendix A.5).

7.6 Leverage, alignment and centrality for the twelve pillars

Table 4 places the latent-structure measures of Section 4, realized importance and the two leverages of Section 6 beside the total effect of each weight, and Figure 7 plots rank leverage against the weights-only total effect.

Table 4. One row per pillar, ordered by rank leverage. σ is the standard deviation of the published pillar scores (13.31 for the composite); ρ^2 the squared correlation with the composite; η^2 the correlation ratio at ten bins; PC1 the first-component loading (Section 4); Strength the network strength (Table 1); L the leverage of (eq4); R the rank leverage $c_j^2(1 - \rho_j^2)$ of (eq5); S_T , weights only, the total effect of the pillar's weight with only the weights varied on the published scores, with its 95 percent bootstrap half-width; and S_T , joint design, the total effect in the design of Section 7.4.

Pillar	σ	ρ^2	η^2	PC1	Strength	L	R	S_T , weights only	S_T , joint design
Personal Freedom	19.91	0.582	0.676	0.75	1.75	0.955	0.935	0.282 $\pm$ 0.008	0.231
Safety and Security	17.54	0.687	0.783	0.82	0.00	0.553	0.544	0.153 $\pm$ 0.005	0.108
Living Conditions	19.75	0.800	0.881	0.89	1.99	0.547	0.440	0.112 $\pm$ 0.004	0.070
Education	19.68	0.829	0.859	0.91	0.47	0.494	0.375	0.116 $\pm$ 0.004	0.082
Governance	17.75	0.854	0.842	0.92	1.32	0.314	0.260	0.076 $\pm$ 0.003	0.065
Social Capital	10.35	0.585	0.612	0.77	0.22	0.416	0.251	0.078 $\pm$ 0.003	0.097
Natural Environment	9.06	0.479	0.562	0.69	0.69	0.521	0.242	0.083 $\pm$ 0.003	0.142
Enterprise Conditions	12.88	0.782	0.806	0.89	1.42	0.225	0.204	0.066 $\pm$ 0.002	0.067
Health	11.17	0.715	0.795	0.85	0.59	0.285	0.201	0.058 $\pm$ 0.002	0.058
Economic Quality	13.43	0.821	0.814	0.91	0.46	0.189	0.182	0.067 $\pm$ 0.002	0.075
Investment Environment	16.74	0.900	0.894	0.95	1.00	0.196	0.158	0.061 $\pm$ 0.002	0.055
Infrastructure and Market Access	15.85	0.890	0.899	0.95	1.00	0.171	0.156	0.060 $\pm$ 0.002	0.056

Rank leverage varied by a factor of 6.0 across the twelve equally weighted pillars, from 0.156 for Infrastructure and Market Access to 0.935 for Personal Freedom. The first-component loading and η^2 ranked the pillars almost identically (Spearman 0.92) and correlated with the joint-design total effect at -0.82 and -0.76 ; both measure alignment, and neither contains the spread factor of (eq5). In the joint design the four highest-loading pillars ranked 12th, 11th, 9th and 6th of the twelve weights by total effect. With only the weights varied on the published scores, however, Education and Living Conditions, loading at 0.91 and 0.89, had the third and fourth largest total effects (0.116 and 0.112), matching the third and fourth largest rank leverages (Living Conditions 0.440, Education 0.375), which come from their spread. A high loading therefore did not preclude a large effect.

Network strength was weakly related to the other measures: its Spearman correlation was 0.15 with the first-component loading, 0.32 with η^2 , 0.11 with rank leverage and -0.23 with the joint-design total effect. Strength measures a different property: it identifies pillars that remain

conditionally connected to others, such as Living Conditions and Personal Freedom, but it does not indicate how much a weight matters, and Safety and Security, with zero strength, had the second largest rank leverage. Expected influence behaved more like an alignment measure (Spearman 0.55 with the loading, 0.61 with η^2 and -0.54 with the joint-design total effect).

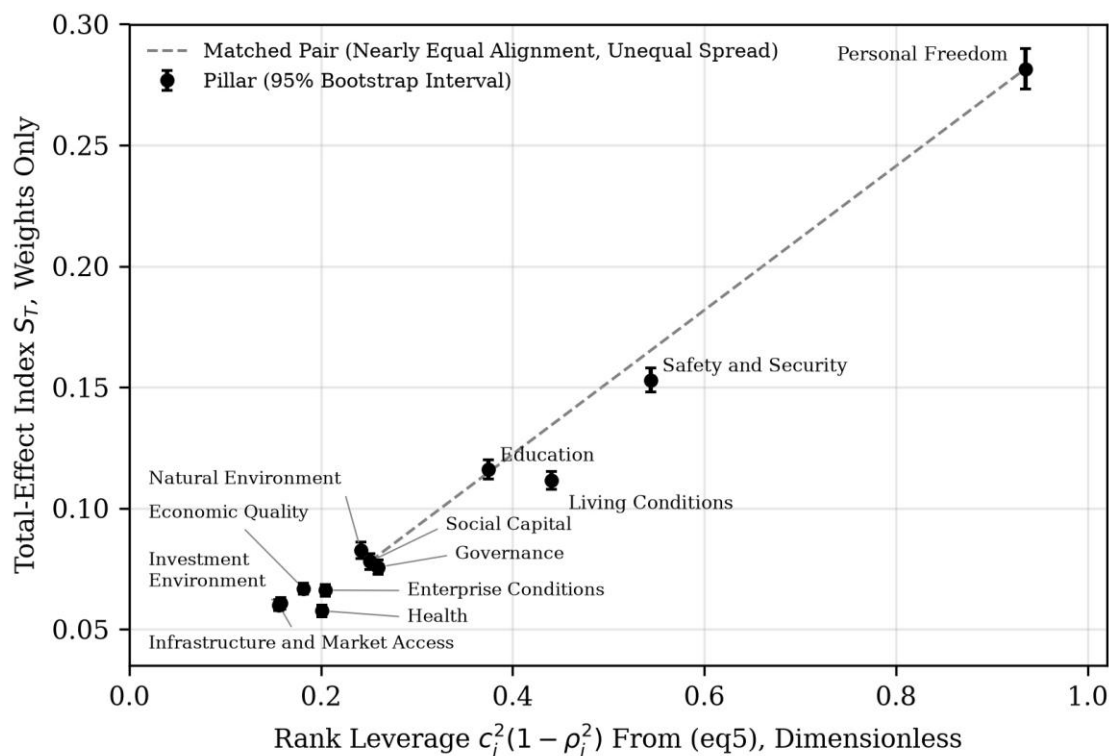


Figure 7. Rank leverage $c_j^2(1 - \rho_j^2)$ from (eq5) against the total-effect index of the same pillar's weight with only the weights varied on the published scores ($N = 8192$), with 95 percent bootstrap intervals. The dashed line joins Personal Freedom and Social Capital, which have nearly equal alignment with the composite and different spreads.

7.7 Country-specific sensitivity profiles

Each country's sensitivity profile was formed from the total-effect indices of its twelve pillar weights, divided by their sum. Because a total effect counts each interaction once for every input involved, these shares measure a weight's influence relative to the other eleven weights, not fractions of the country's rank variance. An unusual pillar profile was defined by regression: each pillar score was regressed on the other eleven across the 167 countries, and a country's local independence on that pillar was the absolute standardized residual, which is large when that score cannot be predicted from the country's other scores.

Averaged across the twelve pillars, the most unusual profiles belonged to Syria (1.51), Equatorial Guinea (1.33), Lesotho (1.33), Myanmar (1.23) and Cambodia (1.23); the most ordinary belonged to the United Kingdom (0.26), Poland (0.33) and the Netherlands (0.34). Per pillar, a country's local independence correlated positively with its total-effect index for that pillar's weight in all twelve cases, at a median Spearman correlation of 0.358, from 0.531 for Social Capital and Safety and Security to 0.099 for Infrastructure and Market Access.

The Personal Freedom weight had the largest share for 46 of the 167 countries, Natural Environment for 30, Social Capital for 22, Safety and Security for 17 and Economic Quality for 13, and each of the remaining seven weights for between three and nine. The same weight was negligible for some countries and influential for others: each of the twelve weights had a share below 0.02 for at least two countries and above 0.20 for at least one. The Personal Freedom weight, the largest in aggregate, ranged from 0.014 for Pakistan to 0.68 for China, and the Living Conditions weight, whose pillar has the third-highest η^2 , from 0.005 for Singapore to 0.42 for Madagascar, the country with the largest local independence on that pillar. Turkmenistan and Belarus, the two widest intervals in Appendix Table A2, were both led by the Personal Freedom weight, at 0.27 and 0.24, and South Sudan by Safety and Security, at 0.29. The 95 percent bootstrap half-widths of the shares quoted in this paragraph, from 200 resamples, were at most 0.014.

Influence also varied with published rank. The median share of the Personal Freedom weight was 0.058 for countries ranked 1 to 33 and 0.074 for those ranked 135 to 167, against 0.169 for ranks 68 to 100, while that of the Living Conditions weight rose from 0.016 in the top tier to 0.070 in the bottom tier. The median rank variance was 7.1 squared places in the top tier and 48.2 in the bottom tier, against 127.9 for ranks 68 to 100.

7.8 Correlates of rank-interval width

Four correlates of the width of the uniform-prior intervals of Section 7.3 were compared. Concentrated sensitivity was measured by normalizing each country's twelve pillar-weight total effects to sum to one and summarizing them with a Herfindahl index (range 0.092 to 0.550, against 0.083 for an even profile); it correlated with interval width at a Spearman correlation of 0.22. Profile unusualness (Section 7.7) correlated with width at 0.48. Local score density, the number of other countries with a published score within one point of a country's own, correlated with width at 0.71, and its tier means of 4.3, 6.3, 11.0, 7.1 and 5.8 countries followed the tier profile of Table 3. Distance from the median published rank correlated with width at -0.70 . Score density and position were more closely associated with interval width than concentrated sensitivity was; the two cannot be separated here, since scores are densest in mid-table.

7.9 Clusters of country sensitivity profiles

The 167 profiles of Section 7.7 were clustered by Ward's method on Euclidean distance, with the number of clusters chosen by silhouette over two to seven; six clusters were selected, with a silhouette of 0.212. When the clustering was repeated on the distinct countries of each of 100 bootstrap samples, its mean adjusted Rand index against the full-sample partition was 0.61. Separation was therefore weak and resampling stability moderate, and the partition is read as a description of a continuum rather than as a set of types.

In five of the six clusters one pillar weight had a mean share above 0.3: Natural Environment (34 countries, 0.328), Personal Freedom (9 countries, 0.558), Social Capital (19 countries, 0.357), Safety and Security (18 countries, 0.326) and Economic Quality (9 countries, 0.336); the

remaining 78 countries formed a diffuse cluster whose largest mean share was 0.169. Members of each of these five clusters had a mean local independence (Section 7.7) of 1.35 to 1.69 on the leading pillar, against 0.62 to 0.75 for the other countries. Apart from Economic Quality, the leading pillars are the four with the lowest ρ^2 in Table 4. No cluster was led by Governance, whose weight had the largest share for 5 of the 167 countries. Against region, eight of the nine countries in the Personal Freedom cluster were in the Middle East and North Africa or East Asia and Pacific, against 37 of all 167 countries; five of the nine in the Economic Quality cluster were Western European, against 20 of 167; and nine of the eighteen in the Safety and Security cluster were Sub-Saharan African, against 49 of 167. The dataset carries no income classification, so published rank tier was used in its place: 43 of the 78 countries in the diffuse cluster ranked 101st or lower, against 67 of all 167.

8. Robustness of the implementation and of the framework

The analyses above vary the implementation of a fixed framework: the twelve pillars, their definitions and the decision to combine them are held constant. Stability under these perturbations therefore shows neither that the framework is uniquely justified nor that it represents prosperity better than an alternative framework would, and no alternative set of dimensions of prosperity was evaluated.

Table 5 rebuilds the 2023 index under alternative constructions of the same twelve pillars and reports the rank shifts in the units of Table 2.

Table 5. Rebuilding the 2023 index under alternative constructions of the same twelve pillars, 167 countries. Shift is the mean absolute change in rank against the published ranking. Reweighting indicates whether some weight vector on the published pillar scores reproduces the ranking exactly.

Construction	Mean shift	Max shift	Spearman	Reweighting
Three domains, equal weight (as published)	0.00	0	1.0000	Yes
Three domains, geometric mean across domains	0.87	8	0.9995	No
Pillars, min–max normalization	1.33	9	0.9991	Yes
Pillars, geometric mean	1.75	12	0.9984	No
First principal component	1.81	10	0.9984	Yes
Pillars, z-score normalization	1.94	11	0.9983	Yes
Pillars, percentile-rank normalization	2.78	14	0.9969	No

Rebuilding at the level of Legatum's three domains reproduced the published ranking exactly, because each domain contains four pillars, so that the mean of the three domain means is the mean of the twelve pillar scores (agreement to 2.8×10^{-14}); the hierarchy differs from flat equal

weighting only when the domains are combined by another rule. The six alternatives moved ranks by 0.87 to 2.78 places on average, the largest single movement being 14 places for Venezuela under percentile-rank normalization, against 2.93 places for the grid prior and 5.68 for the uniform prior (Figure 8). Three of them are reweightings in another form: the min–max, z-score and first-principal-component composites are weighted means of the published pillar scores, with weights of 0.067 to 0.118, 0.060 to 0.132 and 0.053 to 0.108, and each reproduced the ranking of that weighting exactly. No weight vector reproduced the percentile-rank ranking or either geometric-mean ranking, but the pillar-level alternatives are settings of the joint design of Section 5.2, and only the geometric mean across domains lies outside it. Table 5 therefore measures uncertainty in the implementation of the framework, not in the framework itself.

The index maker has also reported sensitivity tests. Its 2023 methodology report compares, on 2019 results, the published ranking with a ranking under equal indicator and element weights and with 1,000 random draws of indicator weights from {0.5, 1, 1.5, 2}, and reports a largest 5th-to-95th-percentile rank range of 12 places (4). The same grid applied to the twelve pillar weights of the 2023 edition gave a median 90 percent interval of 11 places and a largest of 29 places, for Belarus, although the two tests use different editions. In 2026 the Legatum Institute Foundation restructured the index (28): the twelve pillars were replaced by a different tree, the equal-weighted arithmetic mean by power means whose exponent varies with the level of the tree, and equal weights by explicit node weights. The 2026 edition also publishes variants under uniform weights, a uniform aggregation exponent, a uniform normalization function, and their combinations; Table 6 reports the rank movement under all seven.

Table 6. Rank movement in the 2026 Legatum Prosperity Index under the alternative specifications the index publishes for itself, 161 countries. Shift is the mean absolute change in rank against the definitive 2026 ranking.

Alternative specification published by the index	Mean shift	Max shift	Spearman
Uniform normalization function	2.93	13	0.9962
Uniform weights	3.20	15	0.9956
Uniform aggregation exponent	3.75	26	0.9923
Uniform weights and normalization	4.39	21	0.9922
Uniform aggregation exponent and normalization	4.63	26	0.9894
Uniform weights and aggregation exponent	5.34	25	0.9861
All three uniform	5.78	28	0.9840

The index maker's own alternatives moved the 2026 ranking by up to 5.78 places on average, at a Spearman correlation of 0.9840, comparable to the uniform prior for 2023 (5.68 places at 0.9899). The comparison is indicative only: the edition, the structure and the number of countries differ, and the variants are discrete specifications rather than draws from a distribution.

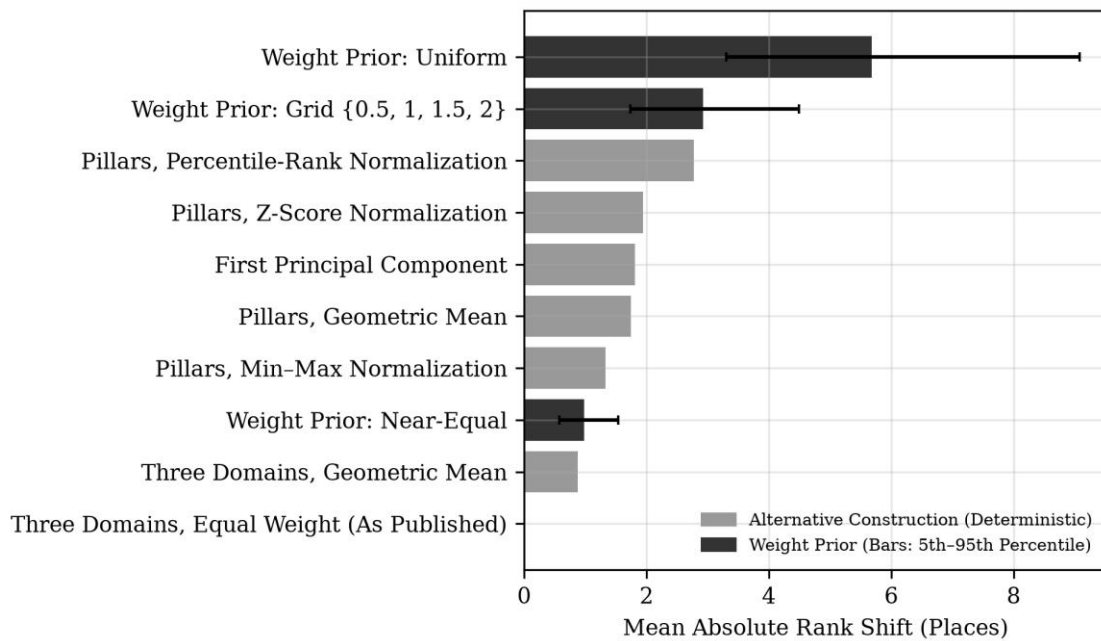


Figure 8. Mean absolute rank shift from the published 2023 ranking under the alternative constructions of Table 5 (grey) and the three weight priors of Table 2 (dark grey, with bars from the 5th to the 95th percentile over draws).

9. Discussion

Equal nominal weights did not give the twelve pillars equal influence on the ranking. Under (eq5) the rank leverage of a weight depends on two properties of its pillar, the share of its variance not shared with the composite and its spread relative to the composite, and in this index it varied by a factor of six (Table 4). Assigning 1/12 to each pillar is therefore an implicit decision about relative influence, and its size can be computed from the pillar scores before any simulation is run. Reporting each pillar's standard deviation and correlation with the composite at sufficient precision, together with the composite's standard deviation, would let users compute these leverages; the 2023 methodology report gives the first two to one decimal place (4).

The same result states when the usual alignment-based importance measures should order the influence of the weights: when pillar spreads are comparable. That held after z-scoring and percentile ranking, where the Spearman correlations of η^2 with the total effects were -0.92 and -0.98 , and not on the published scores (-0.39), where the Personal Freedom weight, whose pillar had the larger spread, had 3.6 times the total effect of the Social Capital weight with only the weights varied, despite nearly identical alignment.

Because mid-table intervals are wide (Table 3), differences of a few places between mid-table countries depend on the weights assumed. Which weight matters most also differs by country (Sections 7.7 and 7.9) and has to be read from that country's own sensitivity profile.

These results concern the construction and robustness of the index, not its validity. No external criterion of prosperity was examined, so neither the stability of the ranking nor the dominance of

one principal component shows that the twelve pillars measure prosperity. The leverage quantities of Section 6 can be computed for any composite index whose final step is a published, replicable weighted arithmetic mean.

10. Limitations

Only the pillar-to-index step was perturbed (Section 3). The selection of the 300 indicators, their imputation, distance-to-frontier normalization and weighting, and their aggregation into 67 elements and twelve pillars were held at their published values. The rank intervals therefore omit the uncertainty of every step below the pillar level, and of the normalization and aggregation choices at this step, and so understate the methodological uncertainty of the index; the Sobol' indices, like the pillar standard deviations and correlations that determine leverage in Section 6, are conditional on those choices. The results also depend on the assumptions of Section 5.4, of which the diffuse uniform prior, the positivity floor before geometric aggregation and the local, arithmetic scope of Section 6 bear most on them.

11. Conclusion

The published ranking of the Legatum Prosperity Index was globally stable, but individual mid-table ranks were not. Under the uniform prior on the pillar weights the median 90 percent rank interval was 22 places, and 36 places for published ranks 68 to 100 (11 and 19 places under the grid prior), and median interval width peaked in the tier where published scores were most densely packed. In the joint design, the weights on Personal Freedom and Natural Environment had the largest total effects. With arithmetic aggregation, the influence of a weight on the ranking followed its rank leverage, $c_j^2(1 - \rho_j^2)$, under every normalization tested, and alignment with the composite ordered the weights closely only where pillar spreads were comparable. Equal nominal weights gave rank leverages that differed by a factor of six. Because only the pillar-to-index step was perturbed, the intervals omit uncertainty below the pillar level and understate the uncertainty of the full methodology. Publishing rank intervals and the leverage of each weight alongside point ranks would show readers which comparisons an index supports.

Conflict of Interest

The authors declare no conflicts of interest.

Data and Code Availability

The analysis uses the file `Dataset_Legatum_Prosperty_Index_2023.xlsx`, published by the Legatum Institute Foundation and not redistributed with the analysis code under its terms of use. The 2023 data file is no longer offered for download on the publisher's website, so it is identified here by content rather than by location: its SHA-256 digest is `8c789bd5ab881c12005f83e9b0c55ae81a68548dd18d3acb9fff9bec0eef631c`, and the analysis reads two of its sheets, Prosperity Index, keyed on `area_code` with columns `score_2023` and `rank_2023`, and Pillars x 12, in long format with columns `area_code`, `pillar_name` and `score_2023`. A reader holding a copy can confirm it is the same data without contacting the

authors: the equal-weighted mean of its twelve pillar scores must reproduce the `off_score` column of `outputs/rank_uncertainty.csv` in the repository, which carries all 167 published scores, to within 10^{-6} . Two secondary results, the repeat computation in Section 6.4 and the published variants in Table 6, use the 2026 edition data file, `Legatum_Prosperty_Index_2026_Data_Sheet.xlsx`, downloaded from the publisher on 8 August 2026, SHA-256 54621ff5d00d0cf4afe6bdbcc47008acff2086cd87cf55c3c8777f819eed1718, which remains available at https://www.prosperityindex.com/downloads/Legatum_Prosperty_Index_2026_Data_Sheet.xlsx; the analysis reads its ranks sheets and the `scores_uni_wghts_uni_p` sheet. The methodology document for the 2023 edition remains available at https://docs.prosperity.com/2116/7756/6547/Measuring_Prosperty.pdf. All analysis code, precomputed outputs, figures and reproducibility tests are available under the MIT License at <https://github.com/RE17-IWD/legatum-prosperty-robustness>. The tests check the two SHA-256 digests, the replication of the published 2023 scores and ranks, the identity (eq4), the equality of the three-domain and twelve-pillar equal-weighted means, and the regeneration of the committed pillar statistics and of the average rank shift under the uniform prior. These links were last accessed on 13 September 2026.

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Author contributions. A.E. designed the study, recovered and verified the 2023 dataset, derived the results of Section 6, directed the sensitivity and country-level analyses of the revision, and wrote the manuscript with the assistance described above. R.E. wrote the analysis code and produced the figures of the original submission, contributed to the design of the study, and reviewed and edited the manuscript. A.E. contributed approximately 70 percent of the work and R.E. approximately 30 percent. Both authors read and approved the final manuscript.

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Appendix A. Formal definitions and distributional details

A.1 Average absolute rank shift. Let r_i^0 be the published rank of country i and $r_i(w)$ its rank under weight vector w . For M draws $w^{(1)}, \dots, w^{(M)}$ from a prior over the simplex, the average absolute rank shift reported as $\bar{R}_S$ in Table 2 is

$$\bar{R}_S = (1 / Mn) \sum_m \sum_i | r_i(w^{(m)}) - r_i^0 | \quad (\text{eq6})$$

with $n = 167$ and $M = 10,000$. It averages over countries as well as draws, so it is not the movement of any particular country; per-country movement is the 90 percent rank interval of Section 7.3.

A.2 Priors over the simplex. The uniform prior is the flat Dirichlet, $w \sim \text{Dir}(1, \dots, 1)$ on the eleven-dimensional simplex, placing equal density on every admissible weight vector including those concentrating almost all weight on one pillar. The near-equal prior is $w \sim \text{Dir}(c, \dots, c)$ with $c = 50$, for which $E[w_j] = 1/12$ and

$$\text{sd}(w_j) = \sqrt{(1/12)(11/12) / (12c + 1)} \approx 0.0113 \quad (\text{eq7})$$

so a weight lies within about three percentage points of $1/12$ with high probability. The grid prior draws each weight independently and uniformly from $\{0.5, 1, 1.5, 2\}$ and renormalizes.

A.3 Hypercube-to-simplex map. The Saltelli design samples the unit hypercube, and the coordinates must be carried onto the simplex. For $u \in (0,1)^{12}$ set $t_j = -\ln u_j$ and $T = \sum_l t_l$. Then

$$w_j(u) = t_j / T \quad (\text{eq8})$$

The t_j are independent standard exponentials, so $w(u)$ is distributed as $\text{Dir}(1, \dots, 1)$ and the map realizes the uniform prior of A.2 exactly. Its derivative is given in Appendix D (eq12).

A.4 Sobol' indices. Write the model output for country i as $Y_i = f(X)$ with $X = (X_1, \dots, X_{14})$ the twelve weight coordinates and the two selector variables, or $X = (X_1, \dots, X_{12})$ the weight coordinates alone in the weights-only designs. The first-order and total-effect indices of input j are

$$S_j = \text{Var}(E[Y | X_j]) / \text{Var}(Y), \quad S_{Tj} = E[\text{Var}(Y | X_{-j})] / \text{Var}(Y) \quad (\text{eq9})$$

estimated for each of the 167 countries separately as described in Section 5.3 and aggregated with weights proportional to that country's rank variance,

$$\bar{S}_j = \sum_i \pi_i S_j^{(i)}, \quad \pi_i = \text{Var}(r_i) / \sum_l \text{Var}(r_l) \quad (\text{eq10})$$

and likewise for the total effects. Since $\pi_i S_{Tj}^{(i)} = E[\text{Var}(r_i | X_{-j})] / \sum_l \text{Var}(r_l)$, the pooled total effect equals $\sum_i E[\text{Var}(r_i | X_{-j})] / \sum_l \text{Var}(r_l)$, the share of the rank variance summed over countries that is attributable to input j ; a country whose rank does not move contributes nothing.

A.5 Correlation ratio. The realized importance of pillar j is the first-order correlation ratio of the composite on that pillar's score, estimated by partitioning the pillar into B quantile bins $b = 1, \dots, B$ and computing

$$\eta_j^2 = \sum_b n_b (\bar{Y}_b - \bar{Y})^2 / \sum_i (Y_i - \bar{Y})^2 \quad (\text{eq11})$$

where n_b is the number of countries in bin b and $\bar{Y}_b$ the mean composite score within it. The estimator makes no linearity assumption. For jointly distributed variables $\eta^2 \geq \rho^2$, with equality only when the conditional expectation of the composite given the pillar is linear. The binned estimate is a between-bin variance share and is not bound by this inequality: it falls below ρ^2 for three pillars at ten bins (Investment Environment, Governance and Economic Quality), for seven at five bins and for none at twenty (Appendix Table A1). Values quoted in the text use $B = 10$; the sensitivity to B is given in Section 5.4 (viii). Figure 6 carries intervals from 1,000 resamples of the 167 countries with replacement, and Figure 4 two standard errors of each tier mean. Because the 167 countries are the population of ranked economies rather than a sample (Section 5.4 (ix)), these are stability diagnostics, showing how much each estimate would move if the set of ranked economies were slightly different, not confidence intervals, and no claim in the text rests on them.

Appendix B. Normalization and aggregation alternatives

The normalization selector takes four values, applied column by column to the 167×12 pillar matrix: identity, which leaves the published scores unchanged; min–max, mapping each pillar to $[0,100]$ by $100(x - \min)/(\max - \min)$; z-score, mapping each pillar to mean zero and unit standard deviation; and percentile rank, the only one of the four that discards the spacing between countries, replacing each score by 100 times its ascending within-pillar rank divided by 167 (values from 0.6 to 100).

The aggregation selector takes two values: weighted arithmetic, $\sum_j w_j \tilde{x}_{ij}$, and weighted geometric, $\exp(\sum_j w_j \ln \tilde{x}_{ij})$, which requires strictly positive inputs. A shift $\delta_j = -\min(\tilde{x}_{\cdot j}) + 10^{-6}$ is added before the logarithm to every column whose minimum is not positive: every column under min–max normalization, which sets each pillar's minimum to zero, and under z-score normalization, which produces negative entries. The published scores and percentile ranks are strictly positive and are not shifted. The lowest-scoring country on each shifted pillar therefore enters the logarithm at 10^{-6} . This floor is an assumption (Section 5.4 (vii)), and Section 7.4 reports the selector total effects under larger floors.

Appendix C. Additional results

Table A1. Realized importance of the twelve pillars. ρ^2 is the squared correlation with the composite; η^2 is the correlation ratio at five, ten and twenty quantile bins; η^2_{-j} is the correlation ratio at ten bins against the composite formed from the other eleven pillars; the share is η^2 at ten bins as a fraction of the total, against a nominal 0.0833.

Pillar	ρ^2	η^2 B=5	η^2 B=10	η^2 B=20	η^2_{-j}	Share
Infrastructure and Market Access	0.890	0.868	0.899	0.915	0.879	0.095
Investment Environment	0.900	0.865	0.894	0.916	0.873	0.095
Living Conditions	0.800	0.856	0.881	0.899	0.854	0.094
Education	0.829	0.816	0.859	0.873	0.825	0.091
Governance	0.854	0.813	0.842	0.864	0.809	0.089
Economic Quality	0.821	0.793	0.814	0.830	0.785	0.086
Enterprise Conditions	0.782	0.779	0.806	0.832	0.776	0.086
Health	0.715	0.781	0.795	0.814	0.771	0.084
Safety and Security	0.687	0.708	0.783	0.798	0.744	0.083
Personal Freedom	0.582	0.640	0.676	0.707	0.609	0.072
Social Capital	0.585	0.516	0.612	0.652	0.575	0.065
Natural Environment	0.479	0.504	0.562	0.570	0.527	0.060

Table A2. The eight widest and four narrowest 90 percent rank intervals under the uniform prior (10,000 weight draws) among the 167 countries; South Africa and Cuba share Colombia's width of 45 places. Width is the 95th minus the 5th percentile of the simulated ranks, in places. The complete per-country file is available in the repository as outputs/rank_uncertainty.csv.

Country	Published rank	5th percentile	95th percentile	Width	Region
Turkmenistan	107	82	137	55	Central and South Asia
Belarus	78	53	105	52	Eastern Europe
Russia	77	54	102	48	Eastern Europe
China	54	44	92	48	East Asia and Pacific
Saudi Arabia	79	58	106	48	Middle East and North Africa
Turkey	95	72	119	47	Middle East and North Africa
Iran	126	106	152	46	Middle East and North Africa
Colombia	81	61	106	45	Latin America and the Caribbean
Cyprus	34	32	36	4	Western Europe
Norway	3	2	6	4	Western Europe
Denmark	1	1	3	2	Western Europe
South Sudan	167	165	167	2	Sub-Saharan Africa

Table A3. Variance-weighted Sobol' indices of country rank for the fourteen inputs of the joint design (Section 5.2), ordered by total effect at N = 8192 (131,072 model evaluations), with the N = 1024 run (16,384 evaluations) alongside. Intervals are 95 percent bootstrap half-widths from 200 resamples.

Input	S (N = 8192)	S _T (N = 8192)	S (N = 1024)	S _T (N = 1024)
Personal Freedom weight	0.191 ± 0.009	0.231 ± 0.007	0.196 ± 0.025	0.229 ± 0.022
Natural Environment weight	0.107 ± 0.006	0.142 ± 0.005	0.111 ± 0.018	0.149 ± 0.017
Safety and Security weight	0.081 ± 0.004	0.108 ± 0.004	0.079 ± 0.012	0.107 ± 0.011
Social Capital weight	0.072 ± 0.004	0.097 ± 0.004	0.071 ± 0.011	0.096 ± 0.010
Aggregation selector	0.047 ± 0.003	0.086 ± 0.003	0.046 ± 0.007	0.086 ± 0.007
Education weight	0.056 ± 0.004	0.082 ± 0.003	0.061 ± 0.011	0.084 ± 0.008
Economic Quality weight	0.052 ± 0.003	0.075 ± 0.003	0.055 ± 0.009	0.076 ± 0.008
Living Conditions weight	0.045 ± 0.004	0.070 ± 0.002	0.042 ± 0.009	0.067 ± 0.006
Enterprise Conditions weight	0.046 ± 0.003	0.067 ± 0.002	0.047 ± 0.008	0.066 ± 0.007
Normalization selector	0.023 ± 0.003	0.067 ± 0.002	0.023 ± 0.006	0.068 ± 0.005
Governance weight	0.044 ± 0.003	0.065 ± 0.003	0.045 ± 0.010	0.064 ± 0.007
Health weight	0.037 ± 0.003	0.058 ± 0.002	0.038 ± 0.008	0.058 ± 0.007

Input	S (N = 8192)	S _T (N = 8192)	S (N = 1024)	S _T (N = 1024)
Infrastructure and Market Access weight	0.032 ± 0.002	0.056 ± 0.002	0.034 ± 0.009	0.056 ± 0.006
Investment Environment weight	0.034 ± 0.002	0.055 ± 0.002	0.035 ± 0.008	0.055 ± 0.006

Appendix D. Derivation of the leverage identities

D.1 The perturbation. Let S be a $167 \times k$ matrix of pillar scores, published or normalized, with columns s_j and $k = 12$; let w lie on the simplex; and let $Y(w) = \sum_j w_j s_j$. The published index is $Y^0 = Y(w^0)$ with $w^0 = (1/k, \dots, 1/k)$. For $w_j = t_j / T$ with $T = \sum_l t_l$ (Appendix A.3), the derivative of (eq8) is

$$\partial w_l / \partial t_j = (\delta_{jl} - w_l) / T \quad (\text{eq12})$$

so increasing t_j moves weight toward pillar j and away from every other pillar in proportion to its current weight, along the direction $(e_j - w)$.

D.2 The change in the composite. A step of size h along that direction from w^0 gives

$$Y(w^0 + h(e_j - w^0)) = Y^0 + h (s_j - Y^0) \quad (\text{eq13})$$

exactly, because the composite is linear in w . The variance of this change across countries is $h^2 \text{Var}(s_j - Y^0)$, and relative to the variance of the composite it is h^2 times the leverage

$$L_j = \text{Var}(s_j - Y^0) / \text{Var}(Y^0) \quad (\text{eq14})$$

D.3 Completing the square. With σ_j and σ_Y the standard deviations of pillar j and of the composite, $c_j = \sigma_j / \sigma_Y$ and ρ_j their correlation,

$$L_j = (\sigma_j^2 - 2 \text{Cov}(s_j, Y^0) + \sigma_Y^2) / \sigma_Y^2 = c_j^2 - 2 \rho_j c_j + 1 \quad (\text{eq15})$$

and adding and subtracting ρ_j^2 gives

$$L_j = (c_j - \rho_j)^2 + (1 - \rho_j^2) \quad (\text{eq16})$$

which is (eq4). On the 2023 data the two sides of (eq16) agreed to 8.3×10^{-16} . With c_j replaced by $w_k R V_k$, where $R V_k$ is the ratio of an indicator's standard deviation to that of the index, (eq16) has the form of the expression under the radical in Formula 3 of Schlossarek, Syrovátka and Vencálek (7), which gives the correlation between an index and the same index recomputed without that indicator.

D.4 The effective-weight constraint. Because $\text{Var}(Y^0) = \text{Cov}(Y^0, \sum_j w_j^0 s_j) = \sum_j w_j^0 \text{Cov}(s_j, Y^0)$, at equal weights $\sigma_Y^2 = (1/k) \sum_j \text{Cov}(s_j, Y^0)$. Writing $a_j = \rho_j c_j = \text{Cov}(s_j, Y^0) / \sigma_Y^2$,

$$(1/k) \sum_j a_j = 1 \quad (\text{eq17})$$

exactly, for any number of pillars k and any score matrix (26). The quantity a_j/k is the share of the composite's variance contributed by pillar j , and a_j is also the slope of the least-squares regression of s_j on Y^0 .

D.5 Rank-neutral and rank-changing parts. Write $s_j = m_j + a_j Y^0 + \varepsilon_j$ for that regression, so that ε_j has mean zero, is uncorrelated with Y^0 and has variance $\sigma_j^2(1 - \rho_j^2)$. Substituting into (eq13),

$$Y^0 + h(s_j - Y^0) = (1 + h(a_j - 1)) Y^0 + h m_j + h \varepsilon_j \quad (\text{eq18})$$

Along the admissible segment $-1/(k-1) \leq h \leq 1$ the factor $1 + h(a_j - 1)$ is positive whenever $0 < a_j < k$, which held for every pillar (a_j ranged from 0.47 to 1.35 on the published scores). The ranks of (eq18) therefore equal the ranks of $Y^0 + [h / (1 + h(a_j - 1))] \varepsilon_j$: the shift $h m_j$ and the rescaling of Y^0 change no rank, and countries are reordered only through ε_j . Since $s_j - Y^0 = m_j + (a_j - 1)Y^0 + \varepsilon_j$ and $\text{Cov}(Y^0, \varepsilon_j) = 0$, (eq14) reduces to (eq5): the first term, $(a_j - 1)^2$, is the rank-neutral rescaling and the second, $c_j^2(1 - \rho_j^2)$, the rank leverage R_j ; on the 2023 data the two sides agreed to 1.1×10^{-16} . If every pillar contributes equally, $a_j = 1$ for all j , the rescaling vanishes, $c_j = 1/\rho_j$ and $R_j = L_j = (1 - \rho_j^2)/\rho_j^2$. If all pillars share one standard deviation, c_j is constant and R_j is proportional to $1 - \rho_j^2$. In both cases rank leverage falls as ρ_j^2 rises, which is the condition discussed in Section 6.3.

Deleting pillar j gives the equal-weighted mean of the other $k - 1$ pillars, $Y_{-j} = (kY^0 - s_j)/(k - 1) = ((k - a_j)Y^0 - m_j - \varepsilon_j)/(k - 1)$. Its correlation with the published index is therefore $(k - a_j) / \sqrt{((k - a_j)^2 + R_j)} = f_j / \sqrt{(f_j^2 + R_j/(k - 1)^2)}$, with $f_j = (k - a_j)/(k - 1)$, so leave-one-out importance is an exact function of a_j and the rank leverage. On the published and the three normalized score matrices the formula matched the directly computed correlations to 4.4×10^{-16} , and one minus the correlation ordered the twelve pillars as rank leverage did at a Spearman correlation of 1.00 on the published scores.